Bioimage analysis 12 years ago (2012)

- ***AlexNet*** is published (2012)
  - Wins the *ImageNet Large Scale Visual Recognition Challenge*
  - Error of 15.3%, more than 10.8 percentage points better than #2
  - Possible due to GPUs
- ***CellProfiler*** has been around for 6 years (2006)
- ***Fiji*** (Fiji is just ImageJ) has been around for 5 years (2007)
- ***scikit-image*** released 3 years ago (2009)
- Bitcoin grows from $5 to $13 (remember Silk Road?) (2012)
- U-Net, GANs, and ***Jupyter*** will appear in 2-3 years (2014/15)
- ***AlphaGo*** will beat Lee Sodol in 4 years (2016)
- ***QuPath*** is still 4 years in the future (2016)

Bioimage analysis 4 years ago (2020)

- CellPose is out (2020)
  - “Cellpose: a generalist algorithm for cellular segmentation”
  - Trained on highly varied images of cells,over 70,000 segmented objects
  - Cells don’t have to be star-shaped
  - Web-platform and Jupyter notebooks
  - https://www.nature.com/articles/s41592-020-01018-x

- ZeroCostDL4Mic available
  - Paper still a year in the future
  - Implementation of common DL technologies to microscopy imaging
  - Relies on GPUs and other infrastructure provided by Google Colab.
  - https://github.com/HenriquesLab/ZeroCostDL4Mic/wiki